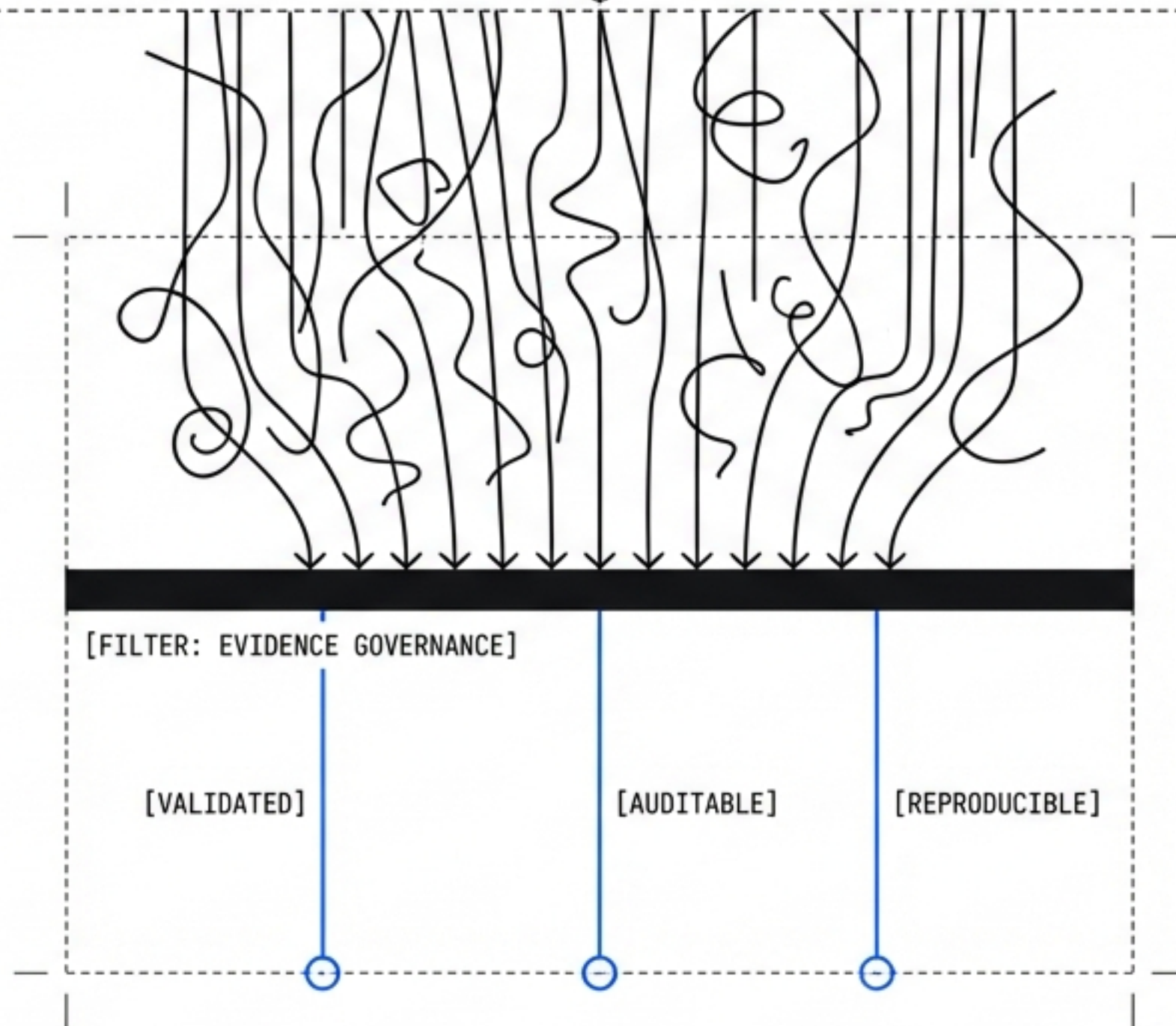


ROOTED LAYERS

AI INSIGHTS GROUNDED ON RESEARCH



The Evidence Governance Pattern

A five-component operational framework for trustworthy LLM-generated analysis.

[CONTEXT: PROTOCOL SECURITY & COMPLIANCE AUDITING]

[EST. 2025 // VERIFICATION OPS]

The Core Insight

Confidence and correctness have decoupled.

THE NARRATIVE (LLM OUTPUT)

Analysis complete. All **[CONFIDENCE: HIGH]** obligations are properly mapped to the specification. The implementation logic strictly follows EIP-1559 pricing mechanics without deviation.



THE DATA (UNDERLYING ARTIFACTS)

[RELEVANCE: LOW/NOISE]

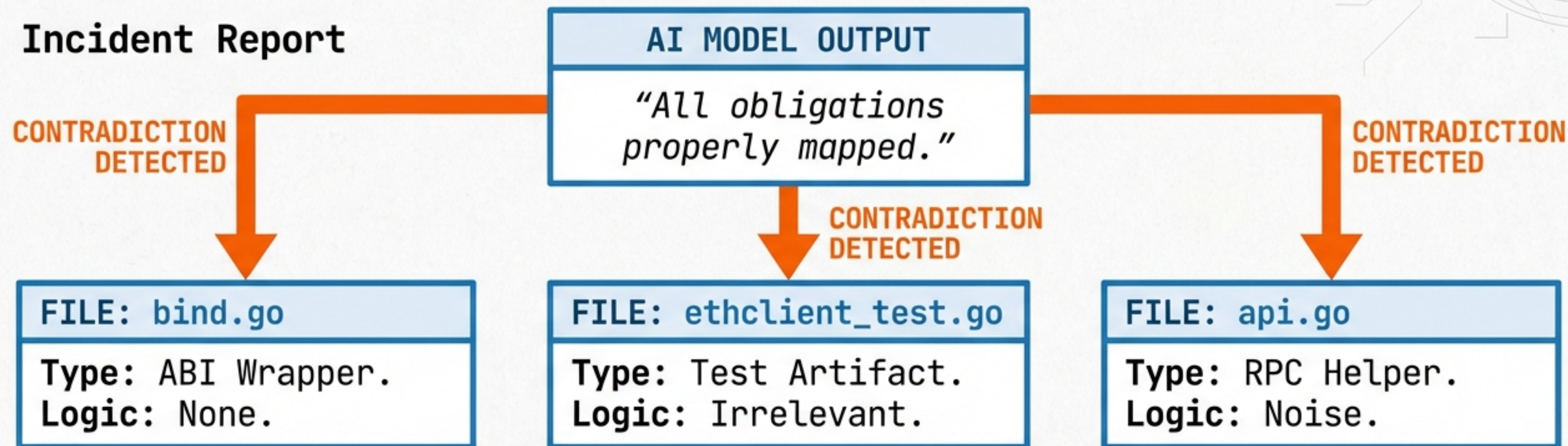
```
{
  files_referenced: [
    'bind.go',
    'ethclient_test.go',
    'internal/ethapi/api.go'
  ]
}
```

The Solution: We adapted Evidence-Based Medicine (EBM) principles to LLM outputs. The 5 Components: 1. Evidence Hierarchy, 2. Scoring Rubric, 3. Contradiction Register, 4. Claim Ledger, 5. Operating Loop.

The Catalyst: Incident CTR-001

A polished narrative silently overrode contradictory evidence.

Incident Report



DIAGNOSIS:

- >> Not a model bug.
- >> A governance failure.
- >> Missing rule: Which artifact to trust when outputs disagree?

The Research Gap

Existing frameworks miss the operational decision.

Capability Benchmarks

HELM, Liang et al.



Measures
standardized
tasks.

Governance Frameworks

NIST AI RMF, Tabassi



Sets
organizational
policy.

THE MISSING LINK: THE OPERATIONAL DECISION

When a specific run
contradicts itself,
what do you **trust**?

Claim Verification

Rasheed et al.

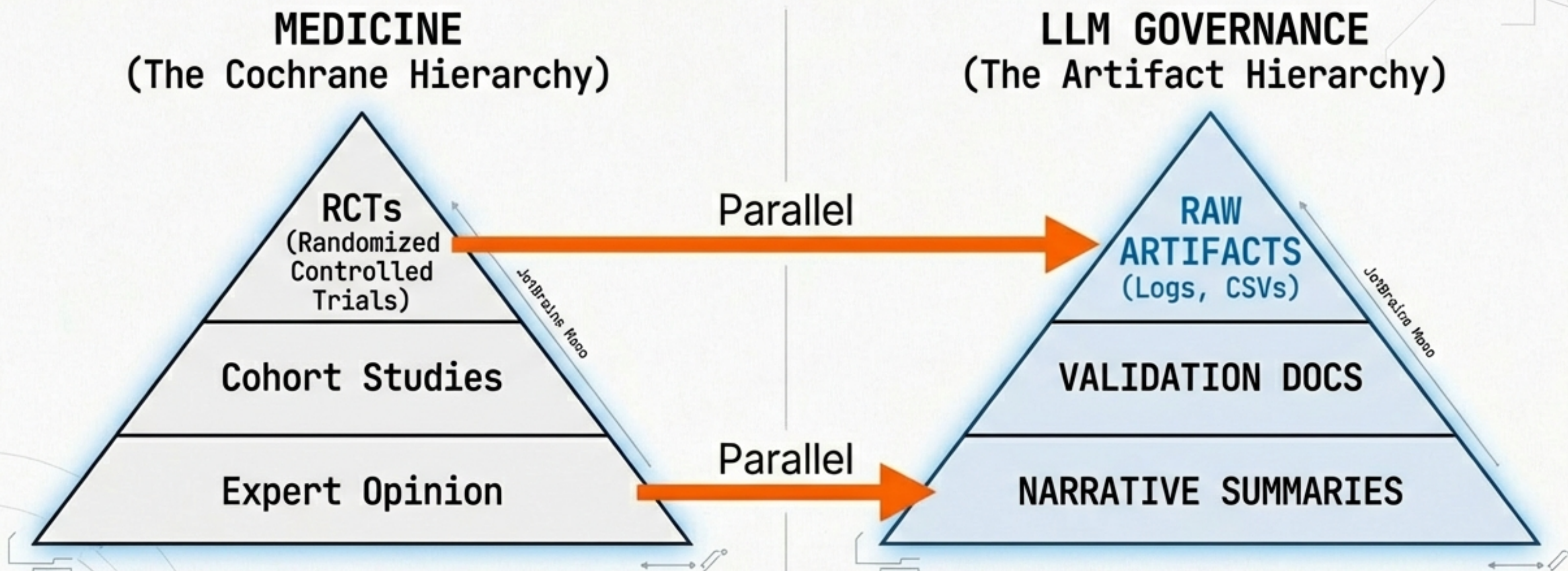


Decomposes
atomic claims.

GOAL: Lower verification burden per claim.

Theoretical Foundation

Adapting Evidence-Based Medicine (Sackett et al., 1996) to LLM Operations.

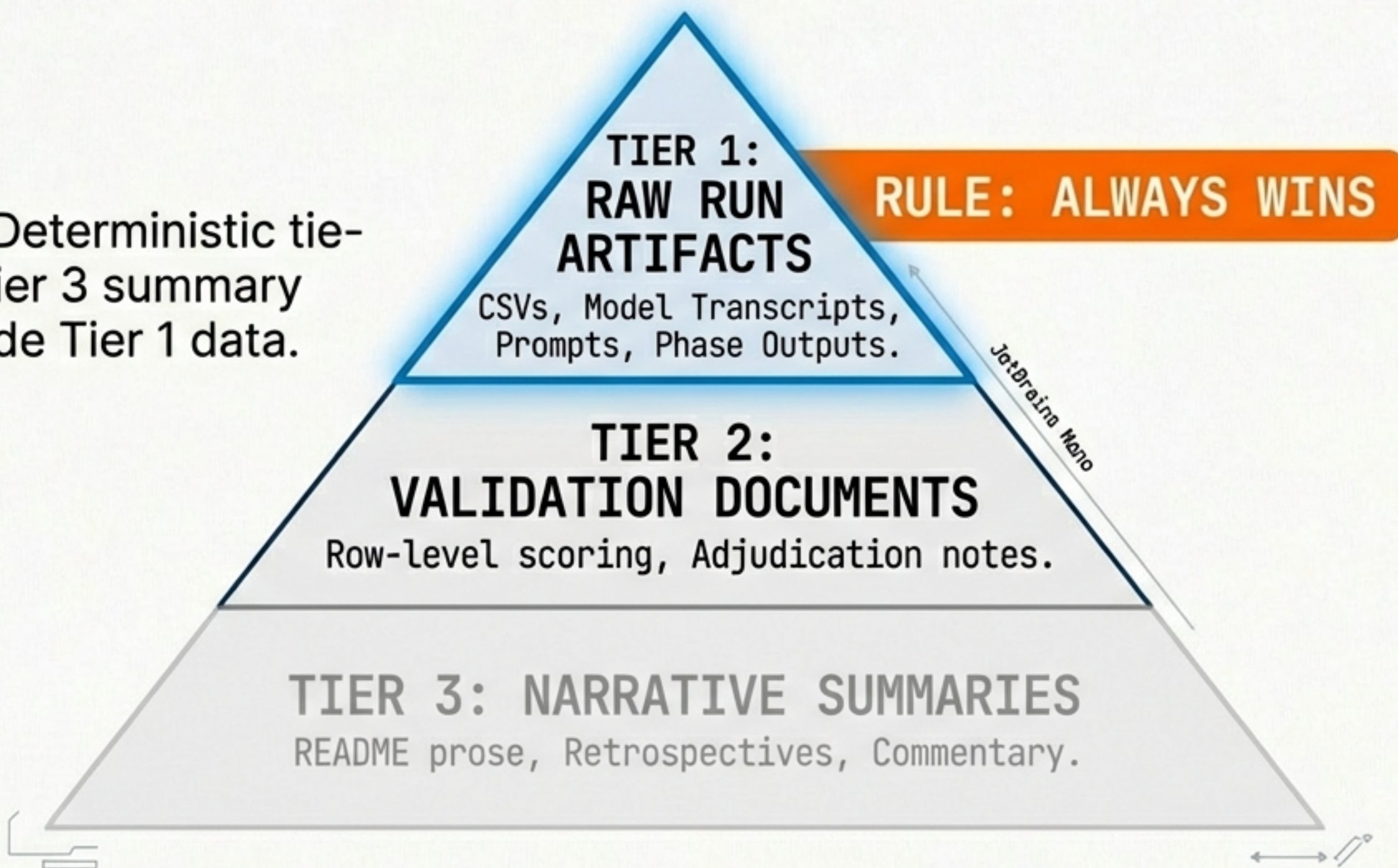


REF: Med-R² (Lu et al., 2025) – Systems must filter lower-level evidence when it conflicts with higher-level sources.

Component 1: The Evidence Hierarchy

Raw Artifacts > Narrative Summaries.

Mechanism: Deterministic tie-breaking. A Tier 3 summary cannot override Tier 1 data.



Component 2: The Scoring Rubric

Moving from Pass/Fail to High-Dimensional Scoring.

Scorecard

JetBrains Mono

DIMENSION	SCORE (0-2)	CRITERIA
Statement Fidelity	[2]	Matches EIP text exactly.
Spec Location Validity	[2]	Code implements logic.
Client Location Validity	[0]	Irrelevant file cited.
Flow Plausibility	[1]	Weak connection.
Evidence Sufficiency	[2]	Full context available.

JetBrains Mono

7/10

PARTIAL / FLAGGED

Outcomes: Valid (≥8), Partial (5-7), Invalid (≤4), Disputed.

Component 3: The Contradiction Register

Disagreements are data, not bugs.

JetBrains Mono

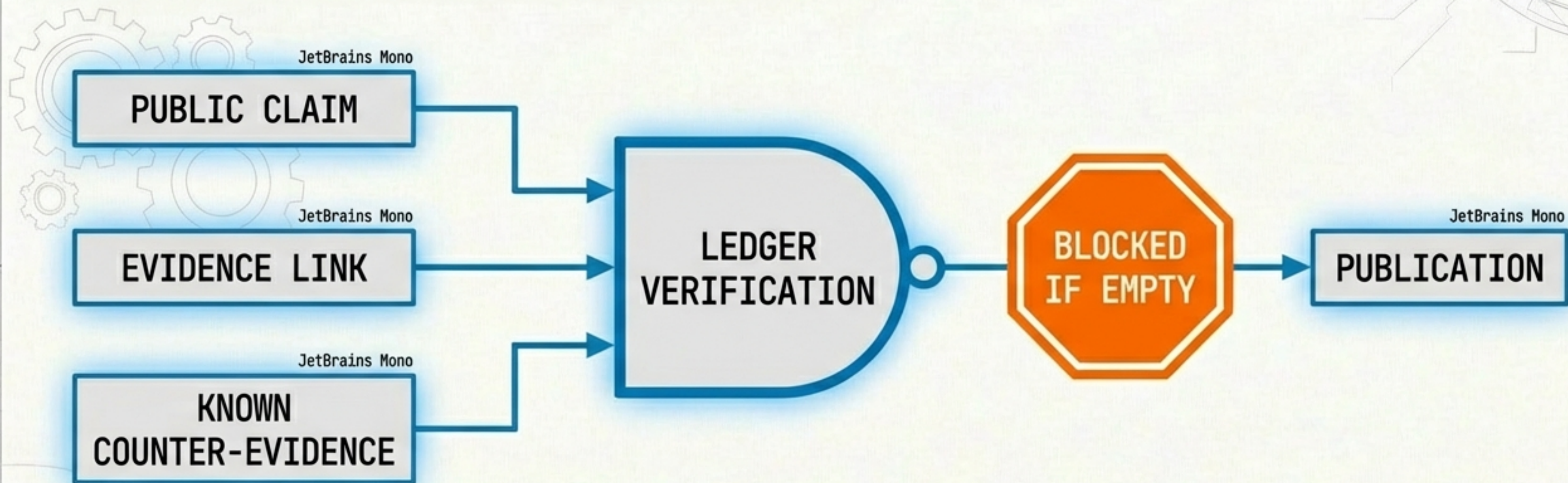
FIELD: ID	FIELD: CONFLICT	FIELD: SOURCES	FIELD: RESOLUTION	FIELD: TIER_PREFERRED
CTR-001	Narrative("Mapped") vs Data("Test File")	summary.md / analysis.csv	Downgrade Narrative	Tier 1 (Data)

Action: Narrative claim downgraded; resolution rule recorded for future runs.

Ref: Rethinking All Evidence (Chen, J. et al., 2025).

Component 4: The Claim Ledger

No external claim without a ledger entry.



Research Backing: From Fluent to Verifiable (Rasheed et al., 2026).

Claims are promoted only when evidence improves; downgraded when it weakens.

Component 5: The Operating Loop

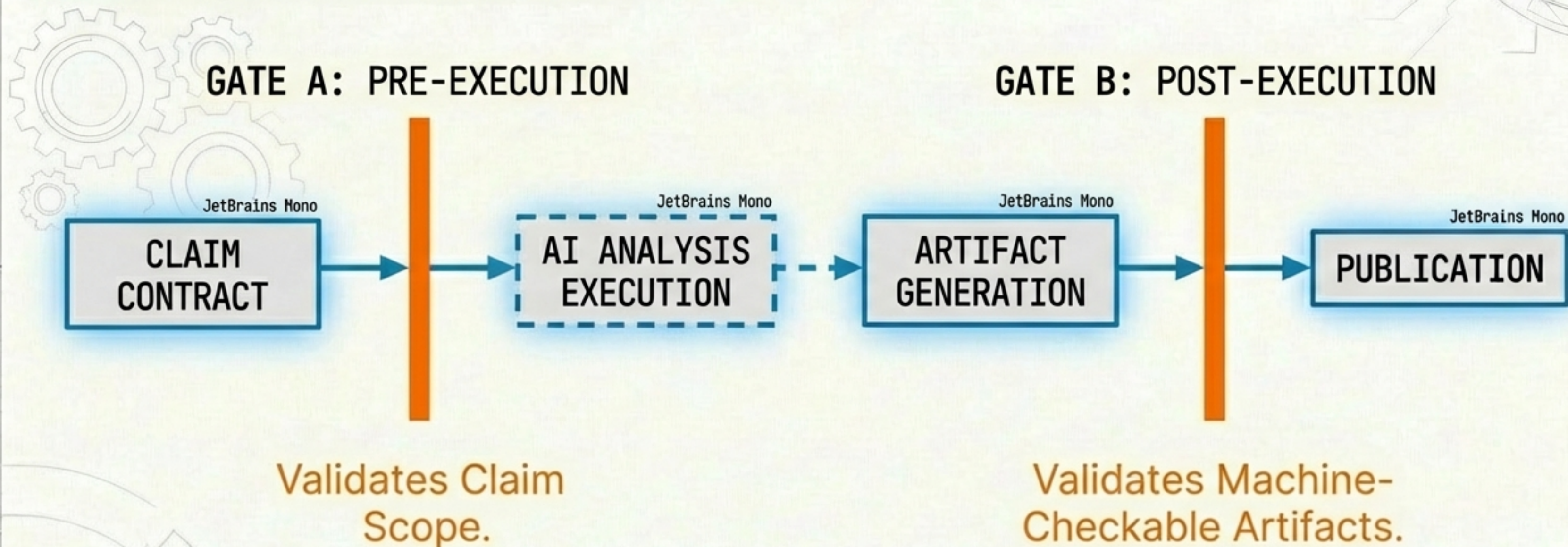
The cycle is productive even when results deteriorate.



SOPBench (Li et al., 2025) - Outcome success does not guarantee procedural compliance.

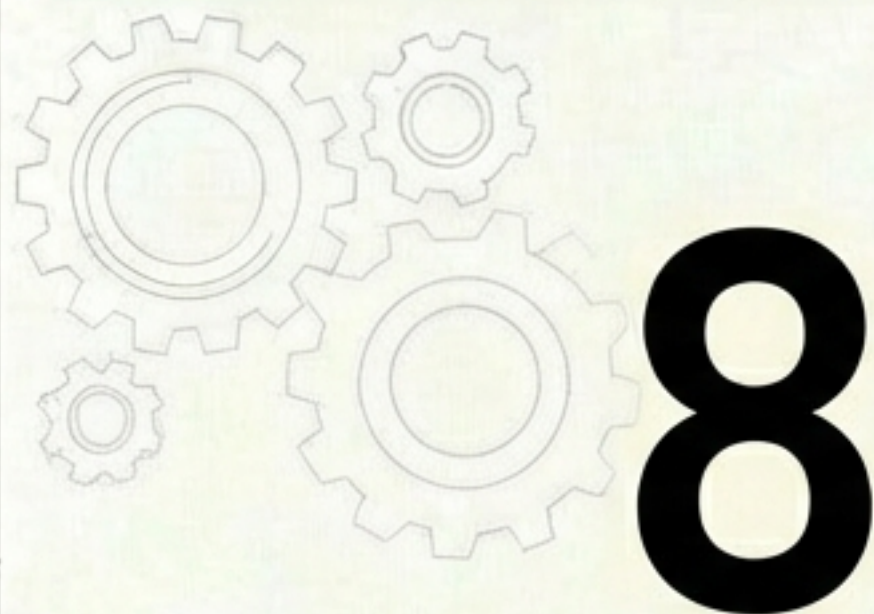
System Architecture: The Dual Gate

Enforcing release discipline.



Results & Impact

Governance exposes where the burden concentrates.



8

CONTRADICTIONS
SURFACED

Across 81 mappings.

3

CLAIMS DOWNGRADED

Prevented false positives.

38.3%

ADJUDICATION BURDEN

Manual review required.

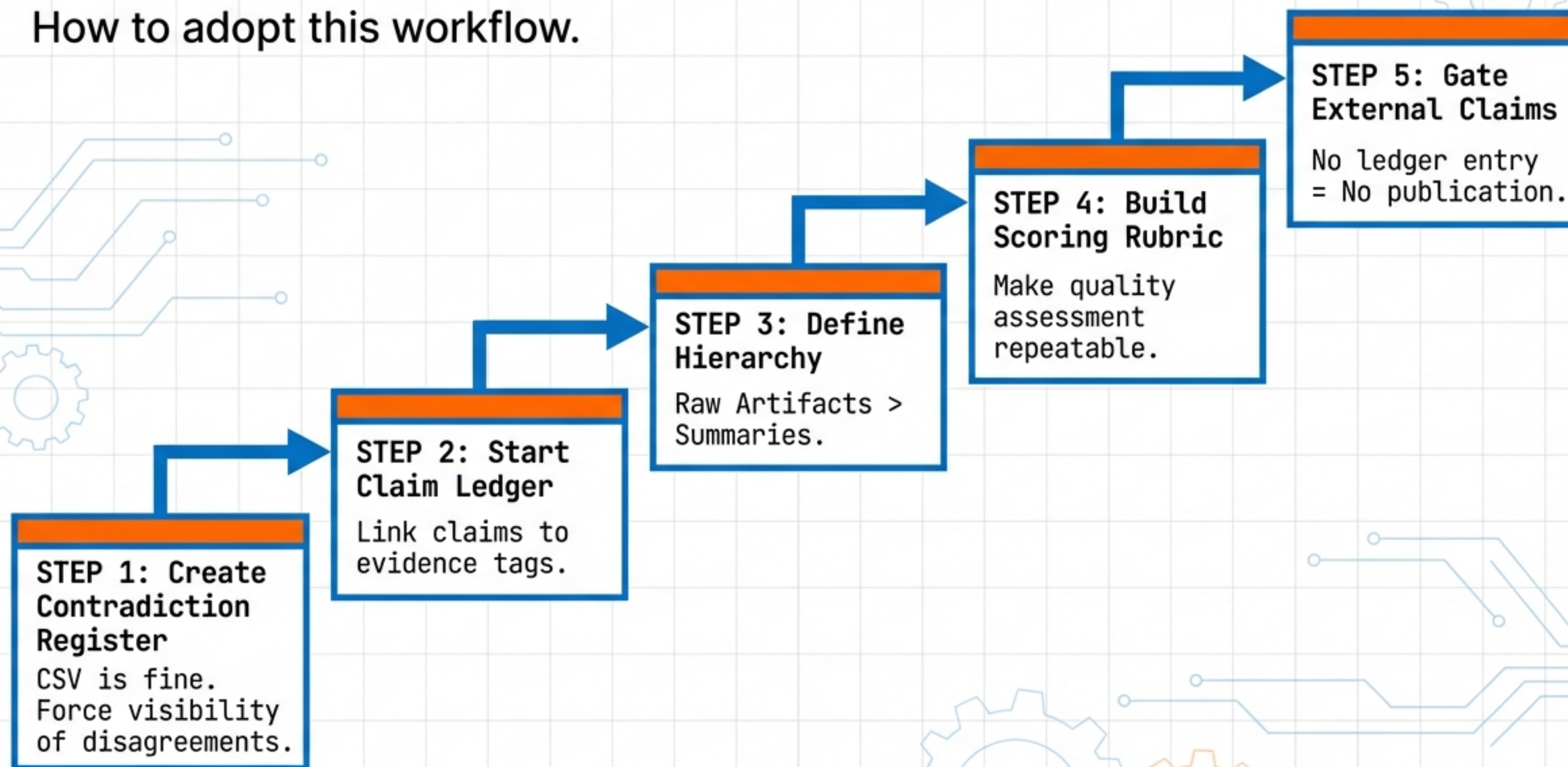
Haiku (Weak Model) - 72.2%

Opus (Strong Model) - 27.8%

Key Insight: Model selection is now a calculated cost decision based on adjudication burden.

Adoption Strategy: The ROI Ladder

How to adopt this workflow.



Limitations

What this framework does NOT solve.



MODEL ACCURACY

Doesn't make the LLM smarter. Just makes errors findable.



HUMAN LOOP

Governance IS review time. It does not remove the expert.



GAMING

Vulnerable to rubber-stamping. Safety cases require the right mindset (Habli et al., 2025).



TRANSFERABILITY

Tested on Ethereum verification. Portable, but not proven.

Resources & References

Rooted Layers | AI Insights Grounded on Research

GITHUB REPO: eip-verify



GITHUB REPO: eip-verify

- Chen, J. et al. (2025). Rethinking All Evidence.
- Chen, R. (2025). Evidence-Bound Autonomous Research.
- Habli, I. et al. (2025). The BIG Argument for AI Safety Cases.
- Ji, Y. et al. (2026). MedRAGChecker.
- Li, Z. et al. (2025). SOPBench.
- Lu, K. et al. (2025). Med-R².
- Rasheed, R. A. et al. (2026). From Fluent to Verifiable.
- Sackett, D. L. et al. (1996). Evidence Based Medicine.
- Simhi, A. et al. (2025). Trust Me, I'm Wrong.

The strongest contribution is a workflow where every trust decision is inspectable, disputable, and updateable.